Chameleon Accounting

User Guide

UK double-entry bookkeeping, Making Tax Digital for VAT, invoices, bank statement import and Open Banking bank feeds

Covers the **Chameleon Accounting MTD** app (desktop and phone) and the command-line tool

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1. About this guide

Chameleon Accounting is desktop software for keeping the books of a UK small company or a self-employed business: a double-entry ledger, a UK chart of accounts, VAT tracking, Profit & Loss and Balance Sheet reports, HMRC-format invoices, Making Tax Digital (MTD) VAT filing, and Open Banking bank feeds.

It runs on your own computer. Your ledger is a single file on your disk. Data leaves your machine only when you tell the software to contact a third party — HMRC when you file a return, or your chosen Open Banking provider when you sync a bank account — and in each case it goes directly from your device to that party.

> **Not tax advice.** Chameleon Accounting is a tool. You, and where applicable your accountant, remain solely responsible for the accuracy, completeness and timeliness of your records and returns, and for reviewing every figure before it is submitted to HMRC. The software does not verify or guarantee acceptance of any submission. See the Terms & Conditions at chameleon-ai-agent-ltd.net/terms.html.

Commands in this guide are shown for the command-line tool, `chameleon_accounting.py`. The desktop application exposes the same features through its Accounting tab. Every command prints a JSON result.

2. The Chameleon Accounting MTD app

Chameleon Accounting MTD is the app for Linux, Windows and macOS desktops and for Android and iPhone. It keeps the same double-entry ledger as the command-line tool and files your VAT return with HMRC. Everything below is done in the app; no commands are needed.

2.1 Sections

Section	What it is for
Dashboard	Cash position across all your bank, payment and cash accounts; recent activity.
Ledger	Every transaction; filter by account (for example one bank or PayPal), search by description or reference.
New entry	A balanced journal entry with VAT codes per line.
Bank feeds	Add your bank or payment accounts and import their statements (2.2).
Reports	Profit & Loss, Balance Sheet, VAT and Review for a chosen period; print, PDF and CSV (2.4).
Invoices	Invoices and quotes (2.3).
AI inbox, Agent	Read a receipt or invoice (PDF or photo) into a draft entry; ask questions about your books (needs your own AI provider key).
VAT / MTD	Connect to HMRC, see open obligations, prepare and submit the return.
Settings	Company details, VAT scheme, backup, export and the database key.

2.2 Bank accounts and statements

In **Bank feeds**, choose *Add bank account* and pick your bank or provider: Barclays, HSBC, Lloyds, Halifax, NatWest, Nationwide, Santander, TSB, Tide, Revolut, Starling, Monzo, Wise, Mettle, ANNA, PayPal, Stripe, SumUp, Square, Zettle and others. Then download a statement from your bank's website as **CSV** or **OFX** (sometimes called Money, Quicken or Microsoft Money format) and choose *Import statement*.

- Each line is shown with a suggested category and VAT rate. Change them if needed, then *Post* or *Ignore*. Nothing reaches your books until you confirm it.
- Importing an overlapping statement again adds nothing twice.

- Bank amounts include VAT; the app splits the VAT out for you.
- A payment that matches an unpaid invoice is suggested as that invoice's payment, so the sale is not counted twice.

2.3 Invoices and quotes

New invoice asks for the client, dates, income account and lines (quantity, price, VAT rate). *Issue invoice* records the sale and its VAT. Tap an invoice to print or save it as PDF, mark it paid (choosing the bank account) or void it; a voided invoice comes off your next VAT return. Quotes do not affect your books until you turn one into an invoice. The PDF carries everything HMRC requires on a VAT invoice and, for a limited company, the Companies Act details.

2.4 Reports, printing and spreadsheets

The print button on **Reports** offers *Print or save as PDF* (P&L, Balance Sheet, VAT and transactions), *This report as CSV* and *Transactions as CSV* for Excel, LibreOffice Calc, Numbers or Google Sheets.

2.5 VAT schemes

Choose your scheme in **Settings**. *Standard* counts sales on the invoice date. *Cash accounting* counts an invoiced sale when the customer pays (VAT Notice 731). *Flat Rate* applies your sector's percentage to your VAT-inclusive turnover (VAT Notice 733); take 1% off in your first year of registration. The VAT tab, the printed reports and the return sent to HMRC all use the scheme you choose.

2.6 Keyboard shortcuts (desktop)

Ctrl on Linux and Windows, ⌘ on macOS. `Ctrl`+`/` (or the keyboard button in the title bar) lists the shortcuts for the screen you are on.

Keys	Does
`Ctrl`+`1` … `9`, `Ctrl`+`0`	Go to Dashboard, Ledger, New entry, Bank feeds, Reports, Invoices, AI inbox, Agent, VAT / MTD; Settings
`Ctrl`+`N`	New entry (new invoice or quote on Invoices)
`Ctrl`+`F`	Search the ledger
`Ctrl`+`S`	Post the entry, save settings, save or issue an invoice
`Ctrl`+`I`	Import a statement (Bank feeds)
`Ctrl`+`P`, `Ctrl`+`E`	Print or PDF, CSV export (Reports)
`Tab`, `Esc`	Move between controls; close a dialog or page

Submitting a VAT return has no shortcut on purpose: it needs the declaration tick and the Submit button.

2.7 Your data, encryption and backups

- **Encrypted.** The app encrypts its whole ledger on your device (SQLCipher v4 format, AES-256). The key is random, made on first use and kept in your device's secure store (Keychain, Android Keystore, the Linux keyring, Windows DPAPI). HMRC tokens and API keys are kept there too.
- **Daily backup.** Each day the app keeps an encrypted copy of the ledger on the device; the last 14 are kept.
- **Nothing is uploaded.** Android cloud backup and iCloud backup are switched off for the app, so your books never go to Google or Apple without you choosing to export them.
- **Export.** *Settings → Backup → Export ledger* saves an **unencrypted** copy where you choose (a USB drive, your own cloud folder, email) so other software can open it. Keep it safe.
- **Your database key.** *Settings → Backup → Show database key* shows the key. Write it down or keep it in a password manager, offline if you can. If your computer's keyring is ever lost (a new computer, a reset), this key is the only way to

open your ledger and its backups.

2.8 Linux: use a standard kernel, not a real-time (PREEMPT_RT) one

Every major Linux distribution installs a standard kernel, which is what the app is built and tested on. Some users install a *full real-time* kernel (`PREEMPT_RT`, often named `-rt`) for audio work. On a 7.2 real-time kernel we have seen the system freeze and crash in the kernel's real-time locking code while running Dart and Flutter software, including this app's tools. A standard kernel of the same version was stable. If you use a real-time kernel and see freezes, switch to your distribution's standard kernel; for low latency, the standard kernel's `preempt=full` boot option is a safer choice. Check with `uname -v`: it shows `PREEMPT_RT` on a real-time kernel. This is unrelated to the app's encryption.

3. Installing and running

Chameleon Accounting ships as part of the Chameleon AI desktop suite for Linux, Windows and macOS. Once installed, the accounting tool is run as:

```
python3 chameleon_accounting.py --help
```

The first run creates the ledger and seeds the UK chart of accounts automatically. Try:

```
python3 chameleon_accounting.py --list-accounts
python3 chameleon_accounting.py --report bs
```

4. Where your data lives

Everything is under `~/.chameleon/accounting/`:

Path	Contents
`<company>/ledger.db`	The double-entry ledger (SQLite).
`<company>/company.json`	Your company details for invoices and reports.
`<company>/hmrc_token.json`	HMRC OAuth tokens (created after you authorise).
`<company>/device_id`	A stable identifier for this device.
`<company>/gocardless_bankdata_*.json`, `truelayer_*.json`	Open Banking tokens and link state.
`api_keys.env`	Your API credentials (mode 600, never committed to version control).
`default_company`	Which company slug to use when `--company` is not given.

Credentials are read from `~/.chameleon/accounting/api_keys.env` (one `KEY=VALUE` per line) or from environment variables of the same name.

5. Companies

The software is multi-company: one ledger per business, each in its own directory. Select a company with `--company <slug>`, set the default once, or use an environment variable:

```
python3 chameleon_accounting.py --set-default-company acme_ltd
export CHAMELEON_ACCOUNTING_COMPANY=acme_ltd     # overrides the default
```

Put the company's legal details in `<company>/company.json` so they appear on invoices and reports:

```
{
  "name": "ACME LTD",
  "company_number": "12345678",
  "address_line1": "1 High Street", "city": "London", "postcode": "EC1A 1AA",
  "country": "United Kingdom",
  "vat_number": "GB123456789",
  "email": "accounts@acme.example", "website": "https://acme.example",
  "bank_name": "", "sort_code": "", "account_number": "",
  "payment_terms_days": 30
}
```

6. The ledger and chart of accounts

A standard UK chart of accounts is seeded on first run — assets (1xxx), liabilities (2xxx), equity (3xxx), income (4xxx), expenses (5xxx–8xxx). Key codes:

Code	Account
1100	Bank: Current Account
1200	Accounts Receivable
2100	Accounts Payable
2210	VAT Control Account
4100–4700	Income (services, product sales, subscriptions, other)
5100–8xxx	Expenses (subcontractors, hardware, cloud, salaries, rent, …)

```
python3 chameleon_accounting.py --list-accounts
```

Every transaction is validated on entry: total debits must equal total credits to the penny, or it is rejected.

7. Recording transactions

7.1 Manual entries

Use `--entry` with a pipe-delimited string: `DATE|DESCRIPTION|MEMO|DEBIT_ACCT|CREDIT_ACCT|AMOUNT[|VAT_CODE]`.

```
python3 chameleon_accounting.py --entry \
  "2026-04-05|Rent|April office rent|7100|1100|102.00|S20"
```

VAT codes: `S20` standard 20%, `R5` reduced 5%, `Z` zero-rated, `E` exempt, `O` outside scope, `EC` reverse charge.

7.2 Bank statement CSV import

The importer auto-detects the layout. Recognised formats include Barclays, HSBC, Lloyds and Monzo business exports, Xero and Wave, PayPal, and generic debit/credit or single-amount CSVs. A file is hashed on import, so the same statement cannot be imported twice.

```
python3 chameleon_accounting.py --import-csv statement.csv --bank-account 1100
```

7.3 OFX / QBO / QFX import

```
python3 chameleon_accounting.py --import-ofx statement.ofx --bank-account 1100
```

Transactions are de-duplicated by their `FITID`.

7.4 Payment processors

Pull settled transactions from a payment processor's API using your own keys:

```
python3 chameleon_accounting.py --gateway-key STRIPE_SECRET_KEY sk_live_...
python3 chameleon_accounting.py --gateway-sync stripe --from 2026-04-01 --to 2026-06-30
```

Supported: `stripe`, `paypal`, `wise`, `gocardless` (Direct Debit), `alipay`, `wechat`. Run `--gateway-setup` for the exact key names each needs. Transactions are de-duplicated by the processor's transaction id.

8. Open Banking bank feeds

Connect a real bank account and let transactions flow in automatically. The connection is made through a **regulated Open Banking provider** — an FCA-authorised Account Information Service Provider (AISP). Two are supported:

- **GoCardless Bank Account Data** (formerly Nordigen) — free tier, ~90 days of history, a few pulls per account per day.
- **TrueLayer** — Data API; sandbox free, live is paid.

> The bank connection runs entirely on your device. Your bank data and bank login are handled by you and the AISP under the AISP's own terms; CHAMELEON-AI-AGENT LTD never receives, stores or has access to them. Open Banking consents expire (typically after 90 days); re-run the link step to renew.

8.1 Choosing a provider

Set the provider per command with `--ob-provider {gocardless|truelayer}`, or once with `OB_PROVIDER=` in `api_keys.env`. Default is `gocardless`.

8.2 GoCardless Bank Account Data

Create a free account at `bankaccountdata.gocardless.com`, then in `api_keys.env`:

```
GOCARDLESS_BANKDATA_SECRET_ID=...
GOCARDLESS_BANKDATA_SECRET_KEY=...
```

8.3 TrueLayer

Create an application in the TrueLayer console (`console.truelayer.com`), register a redirect URI, then:

```
TRUELAYER_CLIENT_ID=...
TRUELAYER_CLIENT_SECRET=...
TRUELAYER_SANDBOX=1        # optional: use the free sandbox environment
```

8.4 Linking a bank

```
# 1. list banks
python3 chameleon_accounting.py --ob-institutions gb

# 2. start the link
#    GoCardless: pass the institution id.  TrueLayer: omit it.
python3 chameleon_accounting.py --ob-link SANDBOXFINANCE_SFIN0000
#    -> open the "link" URL in a browser and authenticate with your bank

# 3. finish the link
#    GoCardless: pass the requisition id.  TrueLayer: pass the ?code= value.
python3 chameleon_accounting.py --ob-poll <requisition-id-or-code>

# 4. confirm
python3 chameleon_accounting.py --ob-accounts
```

8.5 Importing transactions

```
python3 chameleon_accounting.py --ob-import --from 2026-04-01 --to 2026-06-30
```

Options: `--ob-account <id>` to limit to one account, `--ob-ledger-account <code>` to post to a bank account other than 1100. With no dates, the last 90 days are used. Lines are de-duplicated by the bank's own transaction id.

9. Bank reconciliation

Reconciliation checks that the transactions on your ledger's bank account agree with the bank, and matches each bank line to what is already recorded so nothing is double-counted.

9.1 Against the Open Banking feed

```
python3 chameleon_accounting.py --ob-reconcile --from 2026-04-01 --to 2026-06-30
```

This is a dry run — nothing is written. It reports:

- **matched** — bank lines already in the ledger, with how they were matched;
- **unmatched_bank** — bank lines not yet posted (run `--ob-import` to add them);
- **unmatched_ledger** — ledger entries the bank has not cleared (timing differences);
- the bank's own balance vs the ledger balance, and whether they agree.

Match types, best first: *reference* (a shared reference or invoice number), *exact* (same amount and date), *fuzzy_date* (same amount, within a few days), *fuzzy_text* (same amount, close date, similar payee). Widen or narrow the date tolerance with `--ob-reconcile-window DAYS` (default 4).

To import only the genuinely-new lines and skip anything already in the books:

```
python3 chameleon_accounting.py --ob-import --ob-reconcile --from 2026-04-01 --to 2026-06-30
```

9.2 Against a downloaded statement

```
python3 chameleon_accounting.py --reconcile-bank --account 1100 \
  --bank-file statement.csv --statement-balance 12345.67
```

10. Reports

```
python3 chameleon_accounting.py --report pl     # Profit & Loss
python3 chameleon_accounting.py --report bs     # Balance Sheet
python3 chameleon_accounting.py --report vat    # VAT return summary (9 boxes)
python3 chameleon_accounting.py --report txn    # Transaction list
python3 chameleon_accounting.py --forensic      # anomaly scan
```

Date range: `--from` / `--to`, or `--year 2026`, or `--tax-year 2026` (UK 6 April – 5 April). The Balance Sheet also confirms assets = liabilities + equity. Every report is also emitted as structured JSON.

11. Invoices

```
python3 chameleon_accounting.py --invoice new --client "ACME Ltd" \
  --amount 1200.00 --vat S20 --description "AI consulting — April"
python3 chameleon_accounting.py --invoice-list
python3 chameleon_accounting.py --invoice-pay INV-0001
```

Invoices carry every field HMRC Notice 700 requires: a unique sequential number, both VAT numbers, invoice and supply dates, itemised lines with per-line VAT, and net/VAT/gross subtotals. They render as HTML you can print or save as PDF. Company details come from `company.json`.

12. VAT and Making Tax Digital

Chameleon Accounting files VAT returns directly to HMRC under Making Tax Digital, and retrieves your obligations, liabilities and payments.

12.1 One-time setup

1. Register at `developer.service.hmrc.gov.uk` and create an application named, e.g., *Chameleon Accounting*. Subscribe it to **VAT (MTD)** (production and sandbox), **Create Test User** (sandbox) and **Test Fraud Prevention Headers** (sandbox).
2. Put the credentials in `api_keys.env`:

   ```
   python3 chameleon_accounting.py --gateway-key HMRC_CLIENT_ID <client-id>
   python3 chameleon_accounting.py --gateway-key HMRC_CLIENT_SECRET <client-secret>
   python3 chameleon_accounting.py --gateway-key HMRC_VRN <your-9-digit-VRN>
   ```

12.2 Test in the sandbox first

```
# create a sandbox test business (writes its test VRN to api_keys.env)
python3 chameleon_accounting.py --hmrc-create-test-user --hmrc-sandbox

# check the fraud-prevention headers are valid
python3 chameleon_accounting.py --hmrc-check-fraud-headers
```

The `--hmrc-scenario NAME` flag drives HMRC's sandbox test scenarios (e.g. `QUARTERLY_ONE_MET`, `MULTIPLE_LIABILITIES`, `MULTIPLE_PAYMENTS`).

12.3 Authorise

```
python3 chameleon_accounting.py --hmrc-auth --hmrc-sandbox
#   -> open the auth_url, sign in, approve, copy the code shown
python3 chameleon_accounting.py --hmrc-exchange-code <code> --hmrc-sandbox
```

Access lasts about 4 hours and is refreshed automatically; the underlying consent lasts 18 months.

12.4 The quarterly cycle

```
# what needs filing
python3 chameleon_accounting.py --hmrc-obligations --hmrc-sandbox \
  --from 2026-01-01 --to 2026-12-31

# build the return from the ledger and review it
python3 chameleon_accounting.py --hmrc-prepare --hmrc-sandbox \
  --from 2026-04-01 --to 2026-06-30

# dry run (no submission)
python3 chameleon_accounting.py --hmrc-submit --hmrc-sandbox \
  --period-key 26A2 --from 2026-04-01 --to 2026-06-30

# file it for real
python3 chameleon_accounting.py --hmrc-submit --hmrc-sandbox --hmrc-finalize \
  --period-key 26A2 --from 2026-04-01 --to 2026-06-30

# afterwards
python3 chameleon_accounting.py --hmrc-return      --hmrc-sandbox --period-key 26A2
python3 chameleon_accounting.py --hmrc-liabilities --hmrc-sandbox --from 2026-01-01 --to 2026-
12-31
python3 chameleon_accounting.py --hmrc-payments    --hmrc-sandbox --from 2026-01-01 --to 2026-
12-31
```

> **You cannot file a period twice.** If this ledger has already filed the period, the software blocks the submission before it reaches HMRC and tells you the form-bundle number of the earlier filing. Override only if you are certain it is a different filing, with `--hmrc-force`. Once accepted by HMRC, a VAT return cannot be amended — corrections are made in the next period.

12.5 Fraud-prevention headers

HMRC requires every MTD request to carry `Gov-Client-*` / `Gov-Vendor-*` headers that identify the device and connection. Chameleon Accounting generates these from your system (a persistent device id, local IP addresses and their timestamp, MAC addresses, timezone, screen size where available). Validate them against HMRC's checker with `--hmrc-check-fraud-headers`.

12.6 Going live

Once HMRC has granted production credentials, replace `HMRC_CLIENT_ID` / `HMRC_CLIENT_SECRET` in `api_keys.env` with the production pair, set your real `HMRC_VRN`, and drop `--hmrc-sandbox` from every command.

13. Identity checks

```
# UK VAT number: format + check digits (modulus-97 and the 9755 variant)
python3 chameleon_accounting.py --validate-vrn "GB 123 4567 89"

# Companies House: registered name, status, office
python3 chameleon_accounting.py --companies-house 12345678
```

The Companies House lookup needs a free API key from `developer.company-information.service.gov.uk` in `api_keys.env` as `COMPANIES_HOUSE_API_KEY`. Use a key from a *Live* application for real companies; a *Test* key works only against the sandbox (`--companies-house-sandbox`). Neither check proves legal identity — that rests with HMRC's VAT register and the MTD authorisation.

14. Multiple devices and licensing

A licence covers one desktop and one phone, paired together as a seat (typically by scanning a QR code shown by the desktop app). Additional devices may connect and sync as secondary binds. One device at a time is expected to hold MTD filing authority for a set of books; others may view, edit and sync.

> No warranty is offered for data loss, conflicting edits or duplicate/missed filings arising from running the software, or syncing the same books, across more than one device. Review the device log where multiple devices are in use.

```
python3 chameleon_accounting.py --device-log 20
python3 chameleon_accounting.py --device-log-event sync_pull --device-kind phone
```

Every MTD submission attempt — successful or blocked — is written to this log with the device id.

15. Security and backups

- `api_keys.env` and token files are set to mode 600. Keep them off shared drives and out of version control.
- Back up the whole `~/.chameleon/accounting/` directory regularly. The ledger is a single SQLite file per company.
- The **app** encrypts its ledger and daily backups and keeps the key in the device's secure store; keep a copy of the key (*Settings → Backup → Show database key*). The command-line tool's `ledger.db` is not encrypted by the tool: keep it on a full-disk-encrypted drive. See 2.7.
- HMRC and Open Banking tokens live only on your device. Revoke Open Banking access at any time through your bank or the AISP; revoke HMRC access in your HMRC online account.

16. Command reference

Command	Purpose
`--list-accounts`	Show the chart of accounts
`--entry "…"`	Post a manual journal entry
`--import-csv FILE` / `--import-ofx FILE`	Import a bank statement
`--gateway-sync NAME`	Pull from a payment processor
`--ob-provider`, `--ob-institutions`, `--ob-link`, `--ob-poll`, `--ob-accounts`	Set up an Open Banking bank feed
`--ob-import` [`--ob-reconcile`]	Import bank-feed transactions (optionally skipping matched)
`--ob-reconcile`	Reconciliation report (dry run)
`--reconcile-bank --bank-file … --statement-balance …`	Reconcile against a downloaded statement
`--report pl|bs|vat|txn`, `--forensic`	Reports
`--invoice`, `--invoice-pay`, `--invoice-list`	Invoicing
`--hmrc-auth`, `--hmrc-exchange-code`	Authorise with HMRC
`--hmrc-obligations`, `--hmrc-prepare`, `--hmrc-submit` [`--hmrc-finalize`], `--hmrc-return`	File a VAT return
`--hmrc-liabilities`, `--hmrc-payments`	Retrieve VAT liabilities / payments
`--hmrc-check-fraud-headers`	Validate fraud-prevention headers
`--hmrc-create-test-user`, `--hmrc-scenario`, `--hmrc-sandbox`	Sandbox testing
`--validate-vrn`, `--companies-house`	Identity checks
`--company`, `--set-default-company`	Select / set the company
`--device-log`, `--device-log-event`	Device & sync audit log

17. Support

Support covers the software itself — installation, configuration, syncing and features. It is not bookkeeping, accountancy or tax advice, and we do not review your figures or returns.

Contact: `stemsee@chameleon-ai-agent-ltd.net`

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